

Kaplan-Meier Survival App

Output Artefacts: Reference Guide

MoveApps • github.com/meredithspalmer/MoveApps_CoxPropHazard

Introduction

This app implements a Kaplan-Meier (KM) non-parametric survival analysis workflow for animal tracking data loaded from MoveApps. It is designed to handle the realities of telemetry studies: staggered entry (animals collared at different times), right-censoring (animals still alive or lost to follow-up at study end), and group-level comparisons using log-rank tests.

Unlike the Cox Proportional Hazards app, this app does not fit a regression model and does not estimate hazard ratios. Instead, it provides non-parametric survival curves and group comparison tests — making it a useful first step before fitting a Cox model, or a self-contained analysis when covariate adjustment is not required.

The sections below describe every output artefact the app can produce — what it is, what it contains, how to read it, and why it matters. Artefacts are grouped by type: diagnostic summaries, core KM outputs, and group comparison outputs.

Note on optional features: Several plots support two user-controlled options: `add_cis` (add 95% confidence interval ribbons) and `zoom_to_plot` (zoom the y-axis to the observed data range rather than displaying the full axis). When zoom is enabled, an annotation is added to the plot indicating the displayed range.

Note on singleton removal: Groups with only one individual ($N=1$) are automatically detected and removed from KM comparison plots and log-rank tests. A warning is logged indicating which groups were dropped.

Section 1 — Data Diagnostic Artefacts

tracking_history.png

File	tracking_history.png
Type	Figure (PNG)
Always produced?	No — only when 'Tracking history plots' setting is enabled
Dimensions	Dynamic: scales with number of individuals and study duration

What it is

A horizontal Gantt-style chart showing the monitored period for each tagged individual. Each row is one animal; each grey bar spans its deployment period (from `deploy_on_timestamp` to `deploy_off_timestamp`); a blue point marks the start and a purple point marks the end. Upward-pointing arrows mark gaps in tracking longer than 30 days — periods where the individual was not being monitored (e.g., collar fell off and was later retrieved). The subtitle reports the total individual count, visible deployment count, and location count.

What it tells you

This figure provides an at-a-glance overview of your sample: how many individuals are in the study, when they entered and exited, how evenly they are distributed across the study period, and whether any have data gaps worth noting. When running the analysis with survival years defined, the plot is built from the `yearly_survival` table; otherwise it uses the full `summary_table`.

How to interpret it

- Bars that end before the study period ends (especially with a purple endpoint) represent individuals whose deployment ended early — they may be deaths or censored events.
- Bars that extend to the very end of the study are likely censored survivors.
- Gap arrows indicate interrupted monitoring — these individuals contribute observation time in two separate windows.

Why it is valuable

Survival analyses assume you know when each individual entered and exited the risk period. This figure lets you visually confirm the data cleaning and time-windowing steps worked as intended, and flag any unusual deployment patterns before interpreting model results.

monthly_mortality.png

File	monthly_mortality.png
Type	Figure (PNG)
Always produced?	No — only when 'Monthly mortality plots' setting is enabled

What it is

A calendar heat map showing the number of confirmed mortality events per month per year. Months are columns, years are rows, and cell colour encodes mortality count using a PuBu (blue) palette (darker = more deaths). Months with zero deaths are shown in light grey. The subtitle reports the total mortality count and the full date range of the dataset.

What it tells you

This is a purely descriptive diagnostic. It summarises when deaths are happening in calendar time, revealing seasonality, year-to-year variability, and any suspicious clustering (e.g., a batch of deaths in one month that might reflect a discrete mortality event like a disease outbreak or data entry issue).

How to interpret it

- Consistent dark patches in the same months across years indicate seasonal mortality.
- An isolated very dark cell in one year-month may reflect a mass mortality event or a data entry issue worth investigating.
- Many zero-months at the start or end of the time span reflect the ramp-up and wind-down of the study, not necessarily low mortality risk.

Why it is valuable

If you see strong seasonal clustering here, your results may be sensitive to the choice of study start date or survival year definition. This plot is also useful for detecting data anomalies before fitting any model.

Section 2 — Core KM Artefacts

life_table.csv

File	life_table.csv
Type	Table (CSV)
Always produced?	Yes

What it is

A tabular summary of the Kaplan-Meier survival function evaluated at regular time intervals. Columns are:

- `time_days` — the time point at which the survival function is evaluated.
- `n_risk` — the number of individuals still at risk (alive and not yet censored) at that time.
- `n_event` — the number of deaths that occurred at or just before that time point.
- `survival_prob` — the KM-estimated probability of surviving beyond this time point.
- `std_err` — the standard error of the survival estimate.
- `lower_95` / `upper_95` — the lower and upper bounds of the 95% confidence interval.

How it is computed

The KM life table is evaluated at equally spaced time points determined by the `life_table_days` setting. The number of intervals is `ceiling(max_follow_up / life_table_days)`. When survival years are defined, the life table is computed from `days_at_risk` within each survival year; otherwise it uses `entry_time_days` to `exit_time_days` from the full study period.

How to interpret it

The `survival_prob` column at any given time gives the probability that a typical individual in the study would still be alive at that point. Confidence intervals widen at later time points as fewer individuals remain at risk. When `n_risk` falls to very low numbers (e.g., < 5), the survival estimate becomes unreliable regardless of the p-value.

Why it is valuable

The life table is the primary numerical output of the app. It can be used to extract specific survival probabilities at any time point of interest (e.g., 30-day, 90-day, or annual survival), to construct supplementary tables for publications, or to feed downstream analyses.

annual_survival_by_year.csv

File	annual_survival_by_year.csv
Type	Table (CSV)
Always produced?	No — only when survival years are defined

What it is

A table of KM-estimated annual survival probabilities, one row per survival year. Columns are: Year, AnnualSurv (the KM survival probability at the last observed event in that year), SE (standard error), LCI and UCI (95% confidence interval bounds), and n (total number of individuals entering that survival year).

How it is computed

For each survival year, a separate `survfit` object is fit using `Surv(days_at_risk, mortality_event) ~ 1` on the subset of data for that year. The annual survival estimate is taken as the last survival probability in the KM curve for that year — i.e., the probability of surviving to the end of the survival year given entry at the start.

How to interpret it

Annual survival estimates that are stable across years (similar AnnualSurv values) suggest consistent mortality rates over time. Large year-to-year swings may reflect environmental stochasticity, management interventions, or changes in study composition. Wide confidence intervals in some years indicate low event counts and should be interpreted cautiously.

Why it is valuable

Year-specific survival estimates are commonly required for wildlife management reporting and for assessing whether the proportional hazards assumption is likely to hold across years (if annual survival rates differ dramatically between years, time-varying effects may be important to model explicitly).

km_survival_curve.png

File	km_survival_curve.png
Type	Figure (PNG)
Always produced?	Yes
CI option	95% confidence interval ribbon shown when add_cis is enabled
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled

What it is

The overall (unstratified) Kaplan-Meier survival curve for the entire dataset or the user-defined subset. The x-axis is time in days; the y-axis is survival probability from 0 to 100%. A horizontal-vertical dashed line marks the median survival time. The subtitle reports total N, number of events, and median survival in days. When add_cis is enabled, a shaded ribbon shows the 95% confidence interval. When zoom_to_plot is enabled, the y-axis is restricted to the observed survival range and annotated.

What Kaplan-Meier estimation is

The KM estimator is a non-parametric method that estimates the survival function directly from the data without any model assumptions. At each event time, it multiplies the current survival estimate by the fraction of at-risk individuals who survived that moment. It naturally handles right-censoring: censored individuals are removed from the at-risk set at the time of censoring and do not contribute to subsequent estimates.

How to interpret it

The curve descends from 1.0 (all individuals alive at time zero) toward lower probabilities as deaths accumulate. Each step-down represents one or more deaths. The horizontal dashed line at 50% intersects the curve at the median survival time — the point at which half the cohort had died. If the curve does not descend to 50%, the median is undefined (NA), which is common in well-collared populations with good survival rates. In that case, a specific probability at a defined follow-up time (e.g., one-year survival from the life table) is a more useful summary.

Why it is valuable

This is the most intuitive communication tool the app produces. It provides a complete picture of survival dynamics across the entire follow-up period and is suitable for inclusion in papers, reports, and presentations without further processing.

cumulative_hazard_plot.png

File	cumulative_hazard_plot.png
Type	Figure (PNG)
Always produced?	Yes
CI option	95% confidence interval ribbon always shown
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled

What it is

The Nelson-Aalen cumulative hazard plot for the entire dataset. The x-axis is time in days; the y-axis is cumulative hazard $H(t)$, which is unbounded above zero. A 95% confidence ribbon is shown. The subtitle reports total N and number of events.

How to interpret it

The cumulative hazard $H(t)$ can be interpreted as the expected number of mortality events that would have accumulated for a theoretical individual exposed to the same risk over $[0, t]$. It is related to survival by $S(t) = \exp(-H(t))$. Early steep rises indicate periods of high mortality risk; flat stretches indicate near-zero hazard. Unlike the survival curve, the cumulative hazard is not bounded at 1, making it easier to distinguish mortality patterns at late time points when survival probabilities are already low.

Why it is valuable

The cumulative hazard is more sensitive than the survival curve for detecting non-proportional hazards visually. It is also the scale on which the proportional hazards assumption (if a Cox model is subsequently fit) implies parallel lines between groups. Comparing this plot with the group-stratified `cumhaz_comparison_plot.png` is a useful diagnostic.

Section 3 — Group Comparison Artefacts

The following artefacts are produced when a grouping variable (`group_comparison_individual`) is specified. They compare survival across groups within that variable using log-rank tests and stratified KM curves. These artefacts are not produced when no grouping variable is specified.

logrank_table_statistics.csv

File	logrank_table_statistics.csv
Type	Table (CSV)
Always produced?	Only when a grouping variable is specified

What it is

A summary table of the log-rank test comparing survival across groups. The table contains one row per group plus an overall summary row. Per-group columns are: the grouping variable name, N (events) (sample size and event count combined), Expected events (the number of deaths expected under the null hypothesis of equal survival), and O/E ratio (observed-to-expected deaths ratio). The summary row adds: Chisq (log-rank) (the chi-squared test statistic), df (degrees of freedom), and p-value.

How to interpret it

The log-rank test evaluates the null hypothesis that all groups have the same survival function. A small p-value (typically < 0.05) indicates that at least one group has a significantly different survival

trajectory. The O/E ratio gives a simple measure of relative mortality: a ratio > 1 means more deaths than expected (higher mortality), a ratio < 1 means fewer deaths than expected (lower mortality). Groups with O/E ratios furthest from 1 are driving any significant overall p-value.

Why it is valuable

This table provides the primary inferential result for group comparisons — it answers whether the observed survival differences between groups are statistically distinguishable from chance. It is formatted for direct extraction into manuscript results sections.

km_comparison_curves.png

File	km_comparison_curves.png
Type	Figure (PNG)
Always produced?	Only when a grouping variable is specified
CI option	95% confidence interval ribbons shown when add_cis is enabled
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled
Palette	Okabe-Ito colorblind-friendly palette (up to 10 groups)

What it is

A stratified KM survival curve with one line per group, coloured using the Okabe-Ito colorblind-friendly palette. The subtitle reports sample sizes, event counts, and the log-rank p-value for each group. The legend is positioned at the bottom and wraps to two rows when more than eight groups are present. When add_cis is enabled, shaded ribbons show 95% confidence intervals. When zoom_to_plot is enabled, the y-axis is restricted to the observed survival range and annotated. Groups with N=1 are automatically removed before plotting.

How to interpret it

Each step-down in a line represents one or more deaths in that group. If one group's line descends faster and/or sits consistently lower than another's, that group has higher mortality risk. Groups whose confidence intervals overlap substantially do not show statistically distinguishable differences at those time points. The subtitle p-value is from the overall log-rank test and indicates whether any group differences are statistically significant.

Why it is valuable

The stratified KM plot is the most widely recognised way to visualise survival differences between groups and is expected in virtually all ecological survival analysis publications. The Okabe-Ito palette ensures the figure is accessible to readers with colour vision deficiencies.

cum_hazard_comparison_plot.png

File	cum_hazard_comparison_plot.png
Type	Figure (PNG)
Always produced?	Only when a grouping variable is specified
CI option	95% confidence interval ribbons always shown
Zoom option	y-axis zoomed to observed data range when zoom_to_plot is enabled
Palette	Okabe-Ito colorblind-friendly palette (up to 10 groups)

What it is

Stratified cumulative hazard curves, one per group, on the Nelson-Aalen cumulative hazard scale. The subtitle reports per-group sample sizes, event counts, and the overall log-rank p-value. The legend layout and zoom behaviour match the `km_comparison_curves.png` plot.

How to interpret it

- Parallel, non-crossing lines: the proportional hazards assumption holds for this grouping variable. If a Cox model is subsequently fit, the hazard ratio between any two groups is approximately constant over time.
- Lines that converge or cross: the hazard ratio changes over time, violating the PH assumption. The log-rank result may still be valid, but a Cox HR would represent a weighted average effect.
- Lines that start parallel and then fan apart: group differences become more pronounced later in the study period.

Why it is valuable

The cumulative hazard scale makes the proportional hazards assumption visually testable in a way that is more geometrically obvious than residual-based diagnostics. If you plan to subsequently run the Cox PH app on the same data, this plot provides a useful pre-check of whether the PH assumption is plausible for your grouping variable.

This document was generated from the MoveApps_KM Survival App source code. For questions or error reports, contact mspalmer.zool@gmail.com.